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自控第六次作业

7-2

解: (1)  $e(t) = a^n$  ( $e(n) = a^n$ )

$$\begin{aligned}
 E(z) &= e(0) + e(T)z^{-1} + e(2T)z^{-2} + \dots + e(nT)z^{-n} + \dots \\
 &= a^0 + az^{-1} + a^2z^{-2} + \dots + a^nz^{-n} + \dots \\
 &= \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n = \frac{1 \times (1 - \frac{a^n}{z^n})}{1 - \frac{a}{z}} = \frac{1}{1 - \frac{a}{z}} = \frac{z}{z-a}
 \end{aligned}$$

(2)  $e(t) = t^2 e^{-3t}$ 

$$z[e^{-3t} \cdot t^2] = z(t^2) = \frac{T^2 z(z+1)}{(z-1)^3} = E(z)$$

$$\begin{aligned}
 z[e(t)] &= z[e^{-3t} \cdot t^2] = E(z \cdot e^{3T}) \\
 &= \frac{T^2 z e^{3T} (z e^{3T} + 1)}{(z e^{3T} - 1)^3}
 \end{aligned}$$

(3)  $e(t) = \frac{1}{3!} t^3$ 

$$\begin{aligned}
 E(z) &= \frac{1}{3!} [e(0) + e(T)z^{-1} + \dots + e(nT)z^{-n} + \dots] \\
 &= \frac{1}{3!} [0^3 + T^3 z^{-1} + (2T)^3 z^{-2} + \dots + (nT)^3 z^{-n} + \dots] \\
 &= \frac{T^3}{3!} [0 + z^{-1} + 2^3 z^{-2} + \dots + n^3 z^{-n} + \dots]
 \end{aligned}$$

$$又 \frac{z(z^2 + 4z + 1)}{(z-1)^4} = \frac{z^3 + 4z^2 + z}{z^4 - 4z^3 + 6z^2 - 4z + 1} = z^{-1} + 8z^{-2} + \dots + n^3 z^{-n} + \dots$$

$$所以 E(z) = \frac{T^3}{6} \cdot \frac{z(z^2 + 4z + 1)}{(z-1)^4} = \frac{T^3 z(z^2 + 4z + 1)}{6(z-1)^4}$$

(4)  $E(s) = \frac{s+1}{s^2}$ 

$$E(s) = \frac{1}{s} + \frac{1}{s^2}$$

$$E(z) = \frac{z}{z-1} + \frac{Tz}{(z-1)^2} = \frac{z^2 + (T-1)z}{(z-1)^2}$$

$$(5) E(s) = \frac{1-e^{-s}}{s^2(s+1)} = (1-e^{-s}) \left( \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right)$$

$$E(s) = (1-e^{-s}) \left( \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right)$$

$$\because z = e^{sT} \quad \therefore 1-e^{-s} = 1-z^{-\frac{1}{T}}$$

$$E(z) = (1-z^{-\frac{1}{T}}) \left( \frac{Tz}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-T}} \right)$$

$$= \frac{(T-1)z^2 + z + z^2 e^{-T} - (T+1)ze^{-T}}{(z-1)^2(z-e^{-T})} = (1-z^{-\frac{1}{T}}) \left[ \frac{Tz}{(z-1)^2} - \frac{(1-e^{-T})z}{(z-1)(z-e^{-T})} \right]$$

7-4

解: 1,  $E(z) = \frac{z}{(z+1)(3z^2+1)}$

$$\frac{E(z)}{z} = \frac{1}{(z^2+z)(3z^2+1)} = \frac{1}{(z+1)(3z^2+1)}$$

$$E(z) \cdot z^{n-1} = \frac{z^n}{(z+1)(3z^2+1)}$$

$z = -1$  处留数

$$\text{Res} \left[ \frac{z^n}{(z+1)(3z^2+1)} \right]_{z=-1} = \lim_{z \rightarrow -1} \left[ \frac{z^n}{3z^2+1} \right] = \frac{(-1)^n}{4}$$

$z = \frac{\sqrt{3}}{3}i$  处留数

$$\text{Res} \left[ \frac{z^n}{(z+1)(3z^2+1)} \right]_{z=\frac{\sqrt{3}}{3}i} = \lim_{z \rightarrow \frac{\sqrt{3}}{3}i} \left[ \frac{z^n}{3(z+1)(z+\frac{\sqrt{3}}{3}i)} \right] = \frac{(\frac{\sqrt{3}}{3}i)^n}{2\sqrt{3}i-2}$$

$z = -\frac{\sqrt{3}}{3}i$  处留数

$$\text{Res} \left[ \frac{z^n}{(z+1)(3z^2+1)} \right]_{z=-\frac{\sqrt{3}}{3}i} = \lim_{z \rightarrow -\frac{\sqrt{3}}{3}i} \left[ \frac{z^n}{3(z+1)(z-\frac{\sqrt{3}}{3}i)} \right] = \frac{-(-\frac{\sqrt{3}}{3}i)^n}{2\sqrt{3}i-2}$$

$$e(nT) = \frac{(-1)^n}{4} + \frac{(\frac{\sqrt{3}}{3}i)^n - (-\frac{\sqrt{3}}{3}i)^n}{2\sqrt{3}i-2} = \frac{(-1)^n}{4} + \frac{(-1)^n (\frac{\sqrt{3}}{3}i)^n (-2-2\sqrt{3}i) + (\frac{\sqrt{3}}{3}i)^n (-2+2\sqrt{3}i)}{16}$$

$$= \frac{(-1)^n}{4} + \frac{[(-1)^n - 1] i^n (2+2\sqrt{3}i)}{16 \times 3^{\frac{n}{2}}}$$

$$e^*(t) = \sum_{n=0}^{\infty} e(nT) \delta(t-nT)$$

幂级数法

$$E(z) = \frac{z}{3z^3+3z^2+2z+1} = \frac{1}{3} z^{-2} - \frac{1}{3} z^{-3} + \frac{2}{9} z^{-4} - \frac{2}{9} z^{-5} + \dots$$

$$e^*(t) = \frac{1}{3} \delta(t-2T) - \frac{1}{3} \delta(t-3T) + \frac{2}{9} \delta(t-4T) - \frac{2}{9} \delta(t-5T) + \dots$$

2)  $E(z) = \frac{z}{(z-1)(z+0.5)^2}$

$$z^{n-1} E(z) = \frac{z^n}{(z-1)(z+0.5)^2}$$

$z = 1$ ,  $z = -0.5$  处留数:

$$\text{Res} \left[ \frac{z^n}{(z-1)(z+0.5)^2} \right]_{z=1} = \lim_{z \rightarrow 1} \frac{z^n}{(z+0.5)^2} = \frac{4}{9}$$

$$\text{Res} \left[ \frac{z^n}{(z-1)(z+0.5)^2} \right]_{z=-0.5} = \lim_{z \rightarrow -0.5} \frac{1}{1!} \lim_{z \rightarrow -0.5} \frac{d}{dz} \left[ \frac{z^n}{(z-1)} \right] = (-\frac{1}{2})^n \left( \frac{4}{3}n - \frac{4}{9} \right)$$

$$e(nT) = \frac{4}{9} + (-\frac{1}{2})^n \left( \frac{4}{3}n - \frac{4}{9} \right)$$

$$e^*(t) = \sum_{n=0}^{\infty} \left[ \frac{4}{9} + (-\frac{1}{2})^n \left( \frac{4}{3}n - \frac{4}{9} \right) \right] \delta(t-nT)$$

7-8

解: (1)  $c^*(t+2T) - 6c^*(t+T) + 8c^*(t) = r^*(t)$

$r(t) = 1(t) \quad c^*(0) = c^*(T) = 0$

方程等效为  $c(k+2) - 6c(k+1) + 8c(k) = r(k) = 1(k)$

对各项作 Z 变换

$$Z[c(k+2)] = z^2 C(z) - z^2 c(0) - z c(1) = z^2 C(z)$$

$$Z[6c(k+1)] = 6z C(z) - 6z c(0) = 6z C(z)$$

$$Z[8c(k)] = 8C(z)$$

$$Z[1(k)] = \frac{z}{z-1}$$

$$\text{则} (z^2 - 6z + 8)C(z) = \frac{z}{z-1}$$

$$C(z) = \frac{z}{(z-1)(z^2-6z+8)} = \frac{z}{(z-1)(z-2)(z-4)}$$

$$\text{Res}[z^{n-1}C(z)]_{z=1} = \lim_{z \rightarrow 1} [(z-1)z^{n-1}C(z)] = \frac{1}{3}$$

$$\text{Res}[z^{n-1}C(z)]_{z=2} = \lim_{z \rightarrow 2} [(z-2)z^{n-1}C(z)] = -2^{n-1}$$

$$\text{Res}[z^{n-1}C(z)]_{z=4} = \lim_{z \rightarrow 4} [(z-4)z^{n-1}C(z)] = \frac{4^n}{6}$$

$$\text{则} c(nT) = \frac{1}{3} - 2^{n-1} + \frac{1}{6} \times 4^n$$

$$c^*(t) = \sum_{n=0}^{\infty} \left[ \frac{1}{3} - 2^{n-1} + \frac{1}{6} \times 4^n \right] \delta(t-nT)$$

(2)  $c^*(t+2T) + 2c^*(t+T) + c^*(t) = r^*(t)$

$c(0) = c(T) = 0, \quad r(nT) = n \quad (n=0, 1, 2, \dots)$

方程等效为  $c(k+2) + 2c(k+1) + c(k) = r(k)$

对各项作 Z 变换

$$Z[c(k+2)] = z^2 C(z) - z^2 c(0) - z c(1) = z^2 C(z)$$

$$Z[2c(k+1)] = 2z C(z) - 2z c(0) = 2z C(z)$$

$$Z[c(k)] = C(z)$$

$$Z[r(k)] = \frac{T}{z-1} \cdot \frac{z}{(z-1)^2}$$

$$\text{则} C(z) = \frac{Tz}{(z-1)^2(z^2+2z+1)} = \frac{Tz}{(z-1)^2(z+1)^2}$$

$$c(nT) = \text{Res}[z^{n-1}C(z)]_{z=1} + \text{Res}[z^{n-1}C(z)]_{z=-1}$$

$$= \left[ -\frac{n-1}{4} + (-1)^{n-1} \frac{n-1}{4} \right]$$

$$c^*(t) = \sum_{n=0}^{\infty} \left[ -\frac{n-1}{4} + (-1)^{n-1} \frac{n-1}{4} \right] \delta(t-nT)$$

$r(t) = t$   
 $r(nT) = n$   
 $E(z) = 0 + nTz^{-1} + nTz^{-2} + \dots + \frac{Tz^n}{(z-1)^2}$   
 $E(z) = \frac{Tz}{(z-1)^2}$   
 $= 0 + nTz^{-1} + \dots + nTz^{-n}$   
 $= \frac{Tz}{(z-1)^2}$



$$(3) c(k+3) + 6c(k+2) + 11c(k+1) + 6c(k) = 0$$

$$c(0) = c(1) = 2, \quad c(2) = 0$$

对各项作 Z 变换

$$Z[c(k+3)] = z^3 C(z) - z^3 c(0) - z^2 c(1) - z c(2) = z^3 C(z) - z^3 - z^2$$

$$Z[6c(k+2)] = 6z^2 C(z) - 6z^2 c(0) - 6z c(1) = 6z^2 C(z) - 6z^2 - 6z$$

$$Z[11c(k+1)] = 11z C(z) - 11z c(0) = 11z C(z) - 11z$$

$$Z[6c(k)] = 6C(z)$$

$$C(z) = \frac{\cancel{z^3} + \cancel{7z^2} + \cancel{17z}}{z^3 + 6z^2 + 11z + 6} = \frac{z^3 + 7z^2 + 17z}{z^3 + 6z^2 + 11z + 6}$$

$$\frac{C(z)}{z} = \frac{\cancel{z^2} + \cancel{7z} + \cancel{17}}{\cancel{z^3} + 6\cancel{z^2} + 11\cancel{z} + 6} = \frac{z(z^2 + 7z + 17)}{(z+2)(z^2 + 4z + 3)} = \frac{z(z^2 + 7z + 17)}{(z+2)(z+3)(z+1)}$$

$$c(nT) = \text{Res}[z^{n-1} C(z)]_{z \rightarrow -1} + \text{Res}[z^{n-1} C(z)]_{z \rightarrow -2} + \text{Res}[z^{n-1} C(z)]_{z \rightarrow -3}$$

$$= \frac{11}{2} (-1)^n - 7(-2)^n + \frac{5}{2} (-3)^n$$

$$= (-1)^n \left( \frac{11}{2} - 7 \times 2^n + \frac{5}{2} \times 3^n \right)$$

$$c^*(t) = \sum_{n=0}^{\infty} (-1)^n \left( \frac{11}{2} - 7 \times 2^n + \frac{5}{2} \times 3^n \right) \delta(t - nT)$$

$$(4) c(k+2) + 5c(k+1) + 6c(k) = \cos k \frac{\pi}{2}$$

$$c(0) = c(1) = 0$$

对各项作 Z 变换

$$Z[c(k+2)] = z^2 C(z) - z^2 c(0) - z c(1) = z^2 C(z)$$

$$Z[5c(k+1)] = 5z C(z) - 5z c(0) = 5z C(z)$$

$$Z[6c(k)] = 6C(z)$$

$$Z[\cos k \frac{\pi}{2}] = \frac{z(z - \cos \frac{\pi}{2} T)}{z^2 - 2z \cos \frac{\pi}{2} T + 1} = \frac{z^2}{z^2 + 1} \quad (T=1)$$

$$C(z) = \frac{z(z - \cos \frac{\pi}{2} T)}{(z^2 + 5z + 6)(z^2 - 2z \cos \frac{\pi}{2} T + 1)} = \frac{z^2}{(z+2)(z+3)(z^2+1)}$$

$$c(nT) = \text{Res}[z^{n-1} C(z)]_{z \rightarrow -2} + \text{Res}[z^{n-1} C(z)]_{z \rightarrow -3} + \text{Res}[z^{n-1} C(z)]_{z \rightarrow i} + \text{Res}[z^{n-1} C(z)]_{z \rightarrow -i}$$

$$= \frac{(-2)^{n+1}}{5} - \frac{(-3)^{n+1}}{10} + \frac{i^{n+1}}{-10 + 10i} + \frac{(-i)^{n+1}}{10 - 10i}$$

$$= -\frac{2}{5} (-2)^n + \frac{3}{10} (-3)^n + \frac{1}{10} (\cos \frac{\pi}{2} n - \sin \frac{\pi}{2} n)$$

$$c^*(t) = \sum_{n=0}^{\infty} \left[ -\frac{2}{5} (-2)^n + \frac{3}{10} (-3)^n + \frac{1}{10} (\cos \frac{\pi}{2} n - \sin \frac{\pi}{2} n) \right] \delta(t - nT)$$